



K L Deemed to be University

CSE-1 -- KLVZA

Course Handout

2025-2026, Odd Sem

Course Title	:DISCRETE MATHEMATICS
Course Code	:25MT1002
L-T-P-S Structure	: 2-2-0-0
Pre-requisite	:
Credits	: 4
Course Coordinator	:Maddali Radha Madhavi
Team of Instructors	:
Teaching Associates	:

Syllabus :CO1:Basic Discrete Structures: Sets- Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram, Inclusion-Exclusion Principle, Computer Representation of Sets. Relations - Relations and their Properties, Posets, Lattices, Hasse diagram. CO2:Functions- Basic Concepts, Injective and Bijective Functions, Inverse and Composite Functions, Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function. Logic- Propositional Logic, Propositional Equivalences, Rules of Inferences on propositions, Predicates and Quantifiers, Negation of Quantified Statements.CO3: Proof Methods- Direct Proof, Indirect Proof, Proof by Contradiction, Proof by Contraposition. Counting- The basics of counting, the pigeonhole principle, permutations and combinations.C04:Recurrence relations- solving Linear recurrence relations with constant coefficients, Generating functions, solution of recurrence relations by characteristic roots method and Generating functions with applications. Graph Theory- Introduction of Graphs, representation of Graphs, types of graphs, Graph Isomorphism, Euler and Hamiltonian Graphs, Shortest path, Planar graphs, Graph Coloring.

Text Books :1.Kenneth H. Rosen, Discrete mathematics and its applications, McGraw Hill Publication, Seventh Edition,2022. 2. Bernard Kolman, Robert Busby, Sharon C. Ross, Discrete Mathematical Structures, Sixth Edition Pearson Publications, 2015.

Reference Books :1.Joe L Mott, Abraham Kandel, Theodore P Baker, Discrete Mathematics for Computer Scientists and Mathematicians, Printice Hall of India, Second Edition, 2008. 2. Tremblay J P and Manohar R, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill publishers, 1st edition, 2001,India.

Web Links :NPTEL :: Discrete Mathematics- https://onlinecourses.nptel.ac.in/noc23_cs109/preview & Introduction to Discrete Mathematics for Computer Science Specialization- <https://www.coursera.org/specializations/discrete-mathematics>

MOOCS :Discrete Mathematics- https://onlinecourses.nptel.ac.in/noc23_cs109/preview (NPTEL)& Discrete Mathematics- <https://www.udemy.com/course/discrete-math/?couponCode=PMNVD2025> (Udemy)

Course Rationale :Course rationale to include the following aspects: The Purpose of this course is to compute discrete mathematics which emphasizes Mathematical definitions and proofs as well as applicable methods. It introduces logical thinking skills which enables them to solve lifetime problems. Topics include basic Discrete Structures: Sets, Relations, Functions, Logic and Proof Methods, Counting Techniques, Generating Functions, Recurrence Relations and Graph theory.

Course Objectives :The objective of this course is to develop a strong foundation in discrete mathematical structures essential for computer science. Principles of logic, set theory, relations, and functions, and apply various proof techniques, counting methods, along with solving linear recurrence relations and using generating functions, graph theory to solve computational problems and to model real-world problems.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Apply the fundamental concepts of sets and relations to model mathematical structures such as relations, partially ordered sets, lattices using Hasse diagrams.	PO1	3
CO2	Apply the concepts of functions and logic as well as propositional and predicate logic to model and validate logical statements and computational problems in computer science.	PO1	3
CO3	Apply various proof and fundamental counting principles to solve and justify mathematical and computational problems.	PO1	3
CO4	Apply methods for solving linear recurrence relations to computational problems and utilize graph theory to model real-world structures and their applications.	PO1	3

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3
CO1	3	Btl-1 Recall the types of sets and relations with its terminology	Btl-2 Model the given phenomenon into sets, relations and its properties.	Btl-3 Make use of sets, relations, posets and lattice to study discrete structures.
CO2	3	Btl-1 Recall all types of functions and introduction to Propositional and predicate logic	Btl-2 Illustrate various methods of functions and proving techniques in discrete structures, Propositional Logic and Predicate Logic	Btl-3 Make use of logical equivalences to establish the validity of the statements.
CO3	3	Btl-1 Recall various counting techniques.	Btl-2 Apply proof techniques to solve discrete mathematical problems and Permutations.	Btl-3 Apply proof techniques and fundamental counting principles to solve discrete mathematical problems.
CO4	3	Btl-1 Define the graphs with terminology and its types.	Btl-2 Solve the recurrence relations by suitable technique and Summarize different applications of graphs.	Btl-3 Apply linear recurrence relations and graph theory methods to solve computational problems and model real-world structures.

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Introduction to course handout	---	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
2	CO1	COI-1	Sets: Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram	T[1][121-131]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
3	CO1	COI-2	Inclusion-Exclusion Principle, Computer Representation of Sets.	T[1][134-139]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
4	CO1	COI-2	Relations and types of relations	T[1][459-476]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
5	CO1	COI-3	Relations and their Properties.	T[1] [482-502]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
6	CO1	COI-2	Partial order relations and Posets	T[1][502-520]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
7	CO1	COI-3	Hasse diagram	T[1][503-520]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
8	CO1	COI-3	Lattices	R[2] [378-385]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
9	CO2	COI-1	Basic Concepts of functions, Injective and Bijective Functions.	T[1] [141-150]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
10	CO2	COI-1	Inverse and Composite Functions.	T[1][141-150]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
11	CO2	COI-2	Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.	T[1][151-153]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
12	CO2	COI-2	Propositional Logic.	T[1] [1-17]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
13	CO2	COI-3	Propositional Equivalences	T[2] [57-58]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
14	CO2	COI-3	Rules of Inferences on propositions	T[1] [58-72]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
15	CO2	COI-2	Predicates and Quantifiers, Negation of Quantified Statements	T[1][29-53]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1,Tutorial
16	CO3	COI-2	Direct Proof, Indirect Proof	T[1][73-83]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
17	CO3	COI-3	Proof by Contradiction, Proof by Contraposition.	T[1][73-83]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
18	CO3	COI-1	The basics of counting, the pigeonhole principle	T[1][335-353]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
19	CO3	COI-2	Permutations	T[1][354-356]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
20	CO3	COI-3	Combinations.	T[2][96-99]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
21	CO3	COI-3	Solving Problems on Combinations.	T1[347-353] [355-360]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
22	CO4	COI-2	Linear recurrence relations with constant coefficients.	T[1][391-401]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
23	CO4	COI-3	Solution of recurrence relations by characteristic roots method.	T[1][402-414]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
24	CO4	COI-3	Solution of recurrence relations by Generating functions with applications.	T[1][424-439]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
25	CO4	COI-1	Introduction of Graphs, representation of Graphs, types of graphs.	T[1][527-548]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
26	CO4	COI-3	Graph Isomorphism.	T[1][549-559]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
27	CO4	COI-2	Euler and Hamiltonian Graphs.	T[1][571-584]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial
28	CO4	COI-3	Shortest path, Planar graphs, Graph Coluring.	T[1][585-614]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2,Tutorial

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Introduction to Course handout & Understand the sets and set operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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5	Attendance	1	PPT	--- NOT APPLICABLE ---
35	Introduction to Course handout	1	PPT	--- NOT APPLICABLE ---
10	Summary Of the Session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Understand Sets and Subsets. Power Set, Cartesian Product, Set Operations, Make use of Venn Diagram to solve set theory problems.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explaining the concepts of Sets: Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram	2	Chalk	--- NOT APPLICABLE ---
20	solving problems on Sets: Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Apply Inclusion-Exclusion Principle to solve set theory problems, Classifying Computer Representation of Sets.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Explanation of Inclusion-Exclusion Principle, Computer Representation of Sets.	3	PPT	One minute paper
30	problems on Inclusion-Exclusion Principle, Computer Representation of Sets.	3	Chalk	One minute paper
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 Illustrating Relations and their Properties.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction to Relations	2	PPT	--- NOT APPLICABLE ---
20	problems Relations and types of relations	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5**Session Outcome: 1** Identifying Equivalence Relations and their Properties.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Relations and their Properties.	3	PPT	--- NOT APPLICABLE ---
20	Relations and their Properties.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6**Session Outcome: 1** Constructing Partial order relations and Posets

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Explaining concept of Posets	2	PPT	--- NOT APPLICABLE ---
30	Solving problems on Posets	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

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SESSION NUMBER : 7**Session Outcome: 1** Developing Hasse diagram for Posets

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept of Hasse diagram	3	PPT	Leading question
20	Problems on Hasse diagram	3	Chalk	Leading question
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8**Session Outcome: 1** Identifying Lattices for the given Posets.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explaining the concept of Lattices	2	PPT	--- NOT APPLICABLE ---
20	Solving problems on Lattices	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9**Session Outcome: 2** Classification of functions, Injective and Bijective Functions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Basic Concepts of functions, Injective and Bijective Functions.	2	Chalk	--- NOT APPLICABLE ---
20	solving Problems on functions, Injective and Bijective Functions.	2	Chalk	--- NOT APPLICABLE

5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10**Session Outcome: 2** Classification of Inverse and Composite Functions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept of Inverse and Composite Functions.	2	Chalk	--- NOT APPLICABLE ---
20	Solving problems on Inverse and Composite Functions.	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 11**Session Outcome: 2** Classification of Ceiling Function, Floor Function and Boolean Function.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.	2	Chalk	Immediate feedback
20	Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.	2	Chalk	Immediate feedback
5	Summary of the session	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 12**Session Outcome: 2** Explanation of Propositional Logic.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction to Propositional Logic.	2	Chalk	--- NOT APPLICABLE

20	Problems related to Propositional Logic.	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 13**Session Outcome: 2** Applying Logical rules for Propositional Equivalences

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explaining the Concept of Propositional Equivalences	3	Chalk	Statement- Opinion- Summary
20	Solving problems on Propositional Equivalences	3	Chalk	Statement- Opinion- Summary
5	Summary of the session	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 14**Session Outcome: 2** Make use of Rules of Inferences to solve proposition Logic

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Rules of Inferences on propositions	3	Chalk	--- NOT APPLICABLE ---
20	Rules of Inferences on propositions	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15**Session Outcome: 2** Translate Predicates and Quantifiers, Negation of Quantified Statements

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Predicates and Quantifiers, Negation of Quantified Statements	3	Chalk	--- NOT APPLICABLE ---
20	Predicates and Quantifiers, Negation of Quantified Statements	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 16**Session Outcome: 3** Construction of Direct Proof and Indirect Proof

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explanation about Direct Proof, Indirect Proof	3	Chalk	--- NOT APPLICABLE ---
20	Solving Problems on Direct Proof, Indirect Proof	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17**Session Outcome: 3** Construction of Proof by Contradiction, Proof by Contraposition.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Proof by Contradiction, Proof by Contraposition.	3	Chalk	Brain storming session
20	Proof by Contradiction, Proof by Contraposition.	3	Chalk	Brain storming session
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18**Session Outcome: 3** Explanation of The basics of counting, the pigeonhole principle

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction to The basics of counting, the pigeonhole principle	2	Chalk	--- NOT APPLICABLE ---
20	Solving Problems on The basics of counting, the pigeonhole principle	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 19**Session Outcome: 3** solving problems related to Permutations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explaining the concept of Permutations	2	Chalk	Quiz/Test Questions
20	Solving problems on Permutations	2	Chalk	Quiz/Test Questions
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20**Session Outcome: 3** Solving problems related to Combinations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Explanation on Combinations.	3	Chalk	Quiz/Test Questions
20	Solving Problems on Combinations.	3	Chalk	Quiz/Test Questions
5	Summary of the session	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 3 Solving Problems on Combinations and its properties

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Solving Problems on Combinations.	3	Chalk	--- NOT APPLICABLE ---
20	Solving Problems on Combinations.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 22**Session Outcome: 4** Classification of Linear recurrence relations with constant coefficients.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE ---
20	Linear recurrence relations with constant coefficients	2	Chalk	--- NOT APPLICABLE ---
20	Linear recurrence relations with constant coefficients	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 23**Session Outcome: 4** Developing the Solution of recurrence relations by characteristic roots method.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Solution of recurrence relations by characteristic roots method.	3	Chalk	--- NOT APPLICABLE ---
20	Solution of recurrence relations by characteristic roots method.	3	Chalk	--- NOT APPLICABLE ---

5	Summary of the session	2	Chalk	--- NOT APPLICABLE ---
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SESSION NUMBER : 24

Session Outcome: 4 Applying Generating functions to find Solution of recurrence relations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Solution of recurrence relations by Generating functions with applications.	3	Chalk	--- NOT APPLICABLE ---
20	Problems on Solution of recurrence relations by Generating functions with applications.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

Session Outcome: 4 Summarization of Graphs, representation of Graphs, types of graphs.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction of Graphs, representation of Graphs, types of graphs	2	PPT	--- NOT APPLICABLE ---
20	Solving problems on representation of Graphs, types of graphs	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 26

Session Outcome: 4 Identifying Graph Isomorphism in graphs.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

20	Explaining the concept of Graph Isomorphism	3	Chalk	Problem-Based Learning
20	Solving problems on Graph Isomorphism	3	Chalk	Problem-Based Learning
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 27

Session Outcome: 4 Classification of Euler and Hamiltonian Graphs.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction about Euler and Hamiltonian Graphs	3	Chalk	--- NOT APPLICABLE ---
20	Solving problems on Euler and Hamiltonian Graphs	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 28

Session Outcome: 4 Developing the Shortest path, Planar graphs, Graph Colouring.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concept of Shortest path, Planar graphs, Graph Colouring.	2	Chalk	--- NOT APPLICABLE ---
20	Solving Problems on Shortest path, Planar graphs, Graph Colouring.	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:

List of Experiments supposed to finish in Open Lab Sessions:

Lab session no	List of Experiments	CO-Mapping
1	Sets: Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram, Inclusion-Exclusion Principle, Computer Representation of Sets.	CO1
2	Sets: Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram, Inclusion-Exclusion Principle, Computer Representation of Sets.	CO1
3	Relations: Relations and their Properties.	CO1
4	Relations: Relations and their Properties.	CO1
5	Posets, Hasse diagram and Lattices.	CO1
6	Posets, Hasse diagram and Lattices.	CO1
7	Basic Concepts, Injective and Bijective Functions, Inverse and Composite Functions, Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.	CO2
8	Basic Concepts, Injective and Bijective Functions, Inverse and Composite Functions, Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.	CO2
9	Logic: Propositional Logic, Propositional Equivalences	CO2
10	Logic: Propositional Logic, Propositional Equivalences	CO2
11	Rules of Inferences on propositions, Predicates and Quantifiers, Negation of Quantified Statements	CO2
12	Predicates and Quantifiers, Negation of Quantified Statements	CO2
13	Direct Proof, Indirect Proof, Proof by Contradiction, Proof by Contraposition.	CO3
14	Direct Proof, Indirect Proof, Proof by Contradiction, Proof by Contraposition.	CO3
15	Counting: The basics of counting, the pigeonhole principle, Permutations	CO3
16	Permutations	CO3
17	Combinations.	CO3
18	Combinations and its properties	CO3
19	Linear recurrence relations with constant coefficients, solution of recurrence relations by characteristic roots method, solution of recurrence relations by Generating functions with applications.	CO4
20	solution of recurrence relations by Generating functions with applications.	CO4

Lab session no	List of Experiments	CO-Mapping
21	Introduction of Graphs, representation of Graphs, types of graphs, Graph Isomorphism.	CO4
22	Graph Isomorphism.	CO4
23	Euler and Hamiltonian Graphs, Shortest path, Planar graphs, Graph Coloring.	CO4
24	Shortest path, Planar graphs, Graph Coloring.	CO4

Tutorial Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Understand Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Sets: Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram	1	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Set Operations, Venn Diagram	2	Chalk	One minute paper
5	Summary of the Session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 classifying Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram, Inclusion-Exclusion Principle, Computer Representation of Sets.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice the Problems on Computer Representation of Sets.	2	Chalk	One minute paper
20	Practice the Problems on Inclusion-Exclusion Principle	3	Chalk	One minute paper
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3**Session Outcome: 1** Illustrating Relations and their Properties.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Relations	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on properties of relations	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the Session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4**Session Outcome: 1** Identifying Equivalence Relations and their Properties.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Relations	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Equivalence Relations	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5**Session Outcome: 1** Constructing Posets, Hasse diagram and Lattices.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Posets	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Hasse diagrams	3	Chalk	Leading question

5	Summary of the session	1	Talk	--- NOT APPLICABLE ---
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SESSION NUMBER : 6**Session Outcome: 1** Identifying Lattices for the given Posets.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Lattice and Posets	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Lattice and Posets	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7**Session Outcome: 2** Basic Concepts, Injective and Bijective Functions, Inverse and Composite Functions, Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Injective and Bijective Functions,	1	Chalk	Immediate feedback
20	Practice problems on Inverse and Composite Functions.	2	Chalk	Immediate feedback
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8**Session Outcome: 2** Classification of Inverse and Composite Functions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Problems on of Inverse and Composite Functions.	2	Chalk	--- NOT APPLICABLE

20	Problems on of Inverse and Composite Functions.	2	Chalk	Immediate feedback
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9**Session Outcome: 2** Logic: Propositional Logic, Propositional Equivalences

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Propositional Logic,	1	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Truth tables	2	Chalk	Statement- Opinion- Summary
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10**Session Outcome: 2** Applying Logical rules for Propositional Equivalences

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Rules of Inferences on propositions, Predicates and Quantifiers	3	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Negation of Quantified Statements	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 11**Session Outcome: 2** Rules of Inferences on propositions, Predicates and Quantifiers, Negation of Quantified Statements

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Rules of Inferences on propositions	3	Chalk	--- NOT APPLICABLE ---
20	Practice the problems on Predicates and Quantifiers	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12**Session Outcome: 2** Translate Predicates and Quantifiers, Negation of Quantified Statements

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Predicates and Quantifiers,	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Negation of Quantified Statements	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 13**Session Outcome: 3** Direct Proof, Indirect Proof, Proof by Contradiction, Proof by Contraposition.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Direct and Indirect Proofs	3	Chalk	Brain storming session
20	Practice problems on Proof by contradiction	3	Chalk	Brain storming session
5	Summary of the Session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14**Session Outcome: 3** Construction of Proof by Contradiction, Proof by Contraposition.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Proof by Contradiction	3	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Proof by Contraposition.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 15**Session Outcome: 3** The basics of counting, the pigeonhole principle, Permutations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on The basics of counting	2	Chalk	--- NOT APPLICABLE ---
20	Practice the problems on the pigeonhole principle	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16**Session Outcome: 3** solving problems related to Permutations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Permutations	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Permutations and its applications	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Talk	--- NOT APPLICABLE

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SESSION NUMBER : 17**Session Outcome: 3** Combinations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Combinations	2	Chalk	Quiz/Test Questions
20	Practice problems on properties of Combinations	3	Chalk	Quiz/Test Questions
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18**Session Outcome: 3** Solving Problems on Combinations and its properties

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on combinations	3	Talk	--- NOT APPLICABLE ---
20	Practice problems on combinations and properties	3	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19**Session Outcome: 4** Linear recurrence relations with constant coefficients, solution of recurrence relations by characteristic roots method, solution of recurrence relations by Generating functions with applications.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Linear recurrence relations with constant coefficients	2	Chalk	--- NOT APPLICABLE ---

20	Practice problems on Solutions of Linear recurrence relations with constant coefficients	3	Chalk	Leading question
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 4 Applying Generating functions to find Solution of recurrence relations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on solution of recurrence relations by Generating functions with applications.	3	Chalk	--- NOT APPLICABLE ---
20	Practice problems on solution of recurrence relations by Generating functions with applications.	3	Chalk	Leading question
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 4 Introduction of Graphs, representation of Graphs, types of graphs, Graph Isomorphism

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on representation of Graphs	2	Chalk	--- NOT APPLICABLE ---
20	Practice problems on types of Graphs	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

Session Outcome: 4 Identifying Graph Isomorphism in graphs.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

20	Practice problems on Graph Isomorphism.	3	Chalk	Problem-Based Learning
20	Practice problems on Graph Isomorphism.	3	Chalk	Problem-Based Learning
5	Summary of the session	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 4 Euler and Hamiltonian Graphs, Shortest path, Planar graphs, Graph Coloring.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Euler and Hamiltonian Graphs	3	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Shortest path	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the Session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 4 Developing the Shortest path, Planar graphs, Graph Coloring.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Practice problems on Shortest path.	3	Chalk	--- NOT APPLICABLE ---
20	Practice problems on Planar graphs, Graph Coloring.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	2	Chalk	--- NOT APPLICABLE ---

Practical Course DELIVERY Plan: NO Delivery Plan Exists

Practical Session wise Teaching – Learning Plan

No Session Plans Exists

Skilling Course DELIVERY Plan: NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
1	Weekly Homework Assignments	1	Set theory	Sets and Subsets. Power Set, Cartesian Product, Set Operations, Venn Diagram, Inclusion-Exclusion Principle, Computer Representation of Sets.	CO1
2	Weekly Homework Assignments	1	Relations	Relations and their Properties, Inverse and composition of relations, Problems of Equivalence. relations	CO1
3	Weekly Homework Assignments	1	Posets, Hasse diagram and Lattices.	Problems on Posets, Hasse diagram and Lattices.	CO1
4	Weekly Homework Assignments	2	Functions and Types of Functions.	Problems on Injective and Bijective Functions, Inverse and Composite Functions, Functions for Computer Science: Ceiling Function, Floor Function, Boolean Function.	CO2
5	Weekly Homework Assignments	2	Logic	Propositional Logic, Propositional Equivalences	CO2
6	Weekly Homework Assignments	2	Rules of Inferences on propositions.	Problems on Rules of Inferences on propositions, Predicates and Quantifiers, Negation of Quantified Statements	CO2
7	Weekly Homework Assignments	3	Proof Methods	Problems on Direct Proof, Indirect Proof, Proof by Contradiction, Proof by Contraposition.	CO3
8	Weekly Homework Assignments	3	Counting Methods.	Problems on the basics of counting, the pigeonhole principle, Permutations.	CO3
9	Weekly Homework Assignments	3	Combinations.	Problems on Combinations, Properties of combinations and its applications.	CO3
10	Weekly Homework Assignments	4	Linear recurrence relations with constant coefficients.	Problems on Linear recurrence relations with constant coefficients, solution of recurrence	CO4

				relations by characteristic roots method, solution of recurrence relations by Generating functions with applications.	
11	Weekly Homework Assignments	4	Introduction of Graphs Theory.	Introduction of Graphs, representation of Graphs, types of graphs, Graph Isomorphism.	CO4
12	Weekly Homework Assignments	4	Euler and Hamiltonian Graphs, Shortest path, Planar graphs, Graph Coloring.	Problems on Euler and Hamiltonian Graphs, Shortest path, Planar graphs, Graph Coloring.	CO4

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	V-S34,V-S35	V-S34,V-S35	--	--	---	---	V-S45,V-S48	V-S45,V-S48	- - -
	Tutorial	V-S33,V-S33,V-S36,V-S36	V-S33,V-S33,V-S36,V-S36	V-S17,V-S17,V-S18,V-S18,V-S19,V-S19,V-S20,V-S20	V-S17,V-S17,V-S18,V-S18,V-S19,V-S19,V-S20,V-S20	---	---	V-S46,V-S46,V-S47,V-S47	V-S46,V-S46,V-S47,V-S47	- - -
	Lab	--	--	--	--	---	---	--	--	- - -
	Skilling	--	--	--	--	---	---	--	--	- - -
Tue	Theory	V-S9,V-S10,V-S33,V-S36	V-S9,V-S10,V-S33,V-S36	V-S17,V-S18,V-S19,V-S20	V-S17,V-S18,V-S19,V-S20	V-S7,V-S8	V-S7,V-S8	V-S46,V-S47	V-S46,V-S47	- - -
	Tutorial	V-S11,V-S11,V-S12,V-S12,V-S34,V-S34,V-S35,V-S35	V-S11,V-S11,V-S12,V-S12,V-S34,V-S34,V-S35,V-S35	--	--	V-S5,V-S5,V-S6,V-S6	V-S5,V-S5,V-S6,V-S6	V-S45,V-S45,V-S48,V-S48	V-S45,V-S45,V-S48,V-S48	- - -
	Lab	--	--	--	--	--	--	--	--	- - -
	Skilling	--	--	--	--	--	--	--	--	- - -

Wed	Theory	V-S15,V-S16,V-S27,V-S28	V-S15,V-S16,V-S27,V-S28	--	--	V-S1,V-S2	V-S1,V-S2	V-S49,V-S50,V-S51,V-S52	V-S49,V-S50,V-S51,V-S52	- - -
	Tutorial	V-S13,V-S13,V-S14,V-S14,V-S25,V-S25,V-S26,V-S26	V-S13,V-S13,V-S14,V-S14,V-S25,V-S25,V-S26,V-S26	V-S21,V-S21,V-S22,V-S22,V-S23,V-S23,V-S24,V-S24	V-S21,V-S21,V-S22,V-S22,V-S23,V-S23,V-S24,V-S24	V-S3,V-S3,V-S4,V-S4	V-S3,V-S3,V-S4,V-S4	--	--	- - -
	Lab	--	--	--	--	--	--	--	--	- - -
	Skilling	--	--	--	--	--	--	--	--	- - -
Thu	Theory	V-S13,V-S14	V-S13,V-S14	V-S21,V-S22,V-S23,V-S24,V-S25,V-S26,V-S37,V-S38,V-S39,V-S40	V-S21,V-S22,V-S23,V-S24,V-S25,V-S26,V-S37,V-S38,V-S39,V-S40	---	---	--	--	- - -
	Tutorial	V-S15,V-S15,V-S16,V-S16	V-S15,V-S15,V-S16,V-S16	V-S27,V-S27,V-S28,V-S28	V-S27,V-S27,V-S28,V-S28	---	---	V-S49,V-S49,V-S50,V-S50,V-S51,V-S51,V-S52,V-S52	V-S49,V-S49,V-S50,V-S50,V-S51,V-S51,V-S52,V-S52	- - -
	Lab	--	--	--	--	---	---	--	--	- - -
	Skilling	--	--	--	--	---	---	--	--	- - -
Fri	Theory	V-S5,V-S6,V-S43,V-S44	V-S5,V-S6,V-S43,V-S44	--	--	V-S3,V-S4	V-S3,V-S4	---	---	- - -
	Tutorial	V-S7,V-S7,V-S8,V-S8,V-S41,V-S41,V-S42,V-S42	V-S7,V-S7,V-S8,V-S8,V-S41,V-S41,V-S42,V-S42	V-S37,V-S37,V-S38,V-S38,V-S39,V-S39,V-S40,V-S40	V-S37,V-S37,V-S38,V-S38,V-S39,V-S39,V-S40,V-S40	V-S1,V-S1,V-S2,V-S2	V-S1,V-S1,V-S2,V-S2	---	---	- - -
	Lab	--	--	--	--	--	--	---	---	- -

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	Skilling	--	--	--	--	--	--	---	---	-
Sat	Theory	V-S41,V-S42	V-S41,V-S42	---	---	V-S11,V-S12	V-S11,V-S12	---	---	-
	Tutorial	V-S43,V-S43,V-S44,V-S44	V-S43,V-S43,V-S44,V-S44	---	---	V-S9,V-S9,V-S10,V-S10	V-S9,V-S9,V-S10,V-S10	---	---	-
	Lab	--	--	---	---	--	--	---	---	-
	Skilling	--	--	---	---	--	--	---	---	-
Sun	Theory	--	--	--	--	--	--	--	--	-
	Tutorial	--	--	--	--	--	--	--	--	-
	Lab	--	--	--	--	--	--	--	--	-
	Skilling	--	--	--	--	--	--	--	--	-

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4
End Semester	End Semester Exam	Weightage	40		180	10	10	10	10

Summative Evaluation Total= 40 %		Max Marks	100			25	25	25	25
In Semester Formative Evaluation Total= 26 %	Home Assignment and Textbook	Weightage	5		100	1.25	1.25	1.25	1.25
		Max Marks	100			25	25	25	25
	Tutorial	Weightage	15		100	3.75	3.75	3.75	3.75
		Max Marks	100			25	25	25	25
	ALM	Weightage	6		100	1.5	1.5	1.5	1.5
		Max Marks	100			25	25	25	25
In Semester Summative Evaluation Total= 34 %	Semester in Exam-II	Weightage	17		90			8.5	8.5
		Max Marks	50					25	25
	Semester in Exam-I	Weightage	17		90	8.5	8.5		
		Max Marks	50			25	25		

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course

In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Srinivasa Rao Talagadadeevi	L	14-MA,33-MA,37-MA	-	-	-	-
Srinivasa Rao Talagadadeevi	T	9-B,14-A,18-B,23-B,33-A,37-A	-	-	-	-
Venkata K	L	6-MA,19-MA,25-	-	-	-	-

		MA				
Venkata K	T	6-A,19-A,25-A,36-B,39-B,50-B	-	-	-	-
VENU MADHAV SUNKARA	T	21-B	-	-	-	-
Srinivasa Rao Tagallamudi	L	10-MA,23-MA,43-MA	-	-	-	-
Srinivasa Rao Tagallamudi	T	10-A,14-B,15-B,23-A,33-B,43-A	-	-	-	-
Bhavisetti Srinivasa Kumar	L	13-MA,36-MA,38-MA	-	-	-	-
Bhavisetti Srinivasa Kumar	T	6-B,13-A,19-B,24-B,36-A,38-A	-	-	-	-
Appa Rao Bhogapurapu	L	1-MA,20-MA	-	-	-	-
Appa Rao Bhogapurapu	T	1-A,20-A	-	-	-	-
Venkata Subrahmanyam Sajja	L	9-MA,27-MA,48-MA,50-MA	-	-	-	-
Venkata Subrahmanyam Sajja	T	9-A,20-B,27-A,43-B,48-A,50-A	-	-	-	-
Satish Kumar Deevi	L	8-MA,12-MA,24-MA	-	-	-	-
Satish Kumar Deevi	T	8-A,12-A,24-A,25-B,51-B	-	-	-	-
Eswar Lal Tamma	L	44-MA,46-MA,52-MA	-	-	-	-
Eswar Lal Tamma	T	4-B,13-B,16-B,44-	-	-	-	-

		A,46-A,52-A				
Ragamayi Sistla	L	4-MA,17-MA,28-MA,34-MA	-	-	-	-
Ragamayi Sistla	T	4-A,10-B,17-A,28-A,34-A,47-B	-	-	-	-
Bhanu Venkata Varikuti	L	3-MA,16-MA	-	-	-	-
Bhanu Venkata Varikuti	T	3-A,7-B,16-A,44-B,49-B	-	-	-	-
Maddali Madhavi	L	2-MA,45-MA	-	-	-	-
Maddali Madhavi	T	2-A,45-A	-	-	-	-
Bhargavi Yella	L	15-MA,21-MA,42-MA	-	-	-	-
Bhargavi Yella	T	5-B,15-A,17-B,21-A,40-B,42-A	-	-	-	-
Bindu Pathuri	T	11-B,28-B,37-B,41-B	-	-	-	-
Ankita Tiwari	L	18-MA,26-MA,51-MA	-	-	-	-
Ankita Tiwari	T	1-B,12-B,18-A,26-A,46-B,51-A	-	-	-	-
Vijaya Nalleboyina	L	7-MA,47-MA,49-MA	-	-	-	-
Vijaya Nalleboyina	T	7-A,22-B,47-A,49-A	-	-	-	-
S Ibrahim	L	11-MA,22-MA,41-MA	-	-	-	-

S Ibrahim	T	11-A,22-A,41-A	-	-	-	-
Imran Ali	T	8-B,38-B	-	-	-	-
Sravani Jayanti	T	26-B,34-B,42-B	-	-	-	-
LEELAVATHI REKAPALLI	L	35-MA,40-MA	-	-	-	-
LEELAVATHI REKAPALLI	T	35-A,40-A,48-B	-	-	-	-
Nirakara Pradhan	T	2-B,3-B,52-B	-	-	-	-
SEELAM DEVI	L	5-MA,39-MA	-	-	-	-
SEELAM DEVI	T	5-A,39-A	-	-	-	-
SOMNATH KARMAKAR	T	27-B,35-B,45-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Maddali Radha Madhavi)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of Maths

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal) [object HTMLDivElement]